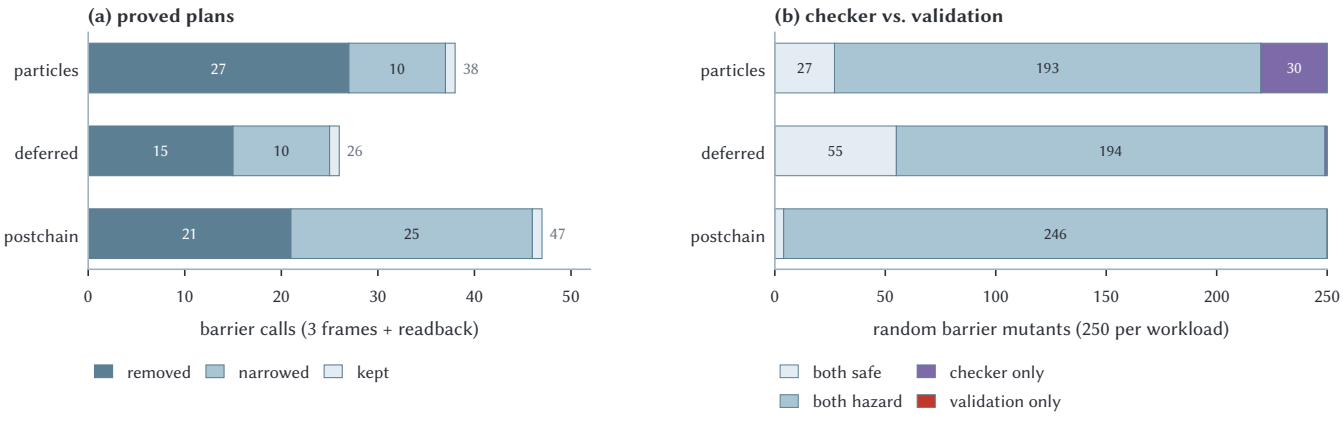
(a) proved plans
particles
deferred
postchain
27
10
38
15
10
26
21
25
47
0
10
20
30
40
50
barrier calls (3 frames + readback)
removed
narrowed
kept
(b) checker vs. validation
particles
deferred
postchain
27
193
30
55
194
246
0
50
100
150
200
250
random barrier mutants (250 per workload)
both safe
checker only
both hazard
validation only